

E082

Linear Algebra for Engineers

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A robotics lab funded by a state grant stores sensor readings in a 3×4 matrix A and a calibration map in a 4×2 matrix B. The lab's lead engineer uses a standing desk. What is the size of AB?

- A. Undefined, because the matrices are different sizes.
- B. 2×3
- C. 4×4
- D. 3×2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A is $5 \times k$ and B is 3×7 . AB is defined. What is k?

- A. 7
- B. 3
- C. 5
- D. Any value works.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An engineer at a water utility serving 80,000 customers must solve $Ax = b$ with $A = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$. The data was collected over two dry summers. What can she conclude?

- A. $\det A = 8$, so there is a unique solution.
- B. A unique solution exists because no entry is zero.
- C. A is singular, so there is no unique solution.
- D. $\det A = 0$, so the only solution is $x = 0$.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

For what value of k is $\begin{bmatrix} k & 3 \\ 4 & 6 \end{bmatrix}$ singular?

- A. $k = 4.5$
- B. $k = -2$
- C. $k = 0$, because any zero entry makes a matrix singular.
- D. $k = 2$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A student at a college known for its engineering co-op program row-reduces an augmented matrix to $\begin{bmatrix} 1 & 2 & | & 3 \\ 0 & 0 & | & 5 \end{bmatrix}$. The system came from a circuit with three resistors. How many solutions does it have?

- A. None
- B. Three, one for each resistor
- C. Infinitely many
- D. Exactly one, $x = 3$

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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6.

A data team at a retail chain with 120 stores builds a 5×8 feature matrix with rank 4. The matrix was exported from a cloud warehouse. What is the dimension of its null space?

- A. 12
- B. 4
- C. 1
- D. 8

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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7. [TRANSFER — reversed stem / different variable]

A 6×10 matrix has a null space of dimension 7. What is its rank?

- A. 3
- B. 17
- C. -1
- D. 7

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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8.

An analyst at a sports-analytics firm with a famous client list has four feature vectors, each with three entries, in $\mathbb{R}^3$. No vector is a multiple of another. The data came from last season. What must be true?

- A. They form a basis of $\mathbb{R}^3$.
- B. Nothing can be concluded without the entries.
- C. The four vectors are linearly dependent.
- D. They are independent, because none is a multiple of another.

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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9.

A graphics engine used by an indie studio in Montreal applies $T(x, y) = (2x, y + 1)$ to every vertex. The studio's last game sold well. Is T a linear transformation?

- A. Yes, because it only scales x by 2.
- B. No, because T does not send the zero vector to zero.
- C. No, because no transformation that multiplies by 2 is linear.
- D. Yes, because each output part is a straight-line formula.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A vibration model for a wind turbine installed offshore last year uses $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$. The turbine's blades are 60 m long. What are A 's eigenvalues?

- A. -5 and -2
- B. 7 and 10
- C. 4 and 3
- D. 5 and 2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A 3×3 matrix has eigenvalues 2, -1 , and 4. What is its determinant?

- A. -8
- B. 5
- C. 0, because a negative eigenvalue makes a matrix singular.
- D. 8

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

An image-compression team at a medical-imaging vendor computes singular values 10, 5, 2, and 1 for a scan matrix. The scan came from a new machine. If they keep only the top two, about what share of the matrix's total energy (sum of squared singular values) is retained?

- A. 50%
- B. About 4%
- C. About 96%
- D. About 83%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A robotics lab ~~funded by a state grant~~ stores sensor readings in a 3×4 matrix A and a calibration map in a 4×2 matrix B. ~~The lab's lead engineer uses a standing desk.~~ What is the size of AB?

Target word: product size **Decides:** Inner sizes 4 and 4 match; outer sizes are 3 and 2.

A	Absolute language	Only the inner dimensions must match.
B	Inverted operation	That is the size of $B^T A^T$, the product in reverse order.
C	Adjacent category	Uses the inner dimensions, which cancel.
D	Correct	$(3 \times 4)(4 \times 2) = 3 \times 2$.

2. Answer B Concept 1 · transfer

A is $5 \times k$ and B is 3×7 . AB is defined. What is k?

Target word: solve for the inner dimension **Decides:** Columns of A must equal rows of B.

A	Wrong stem number	B's column count, which sets the result's width.
B	Correct	k must equal B's row count, 3.
C	Inverted operation	Matches A's rows instead of B's rows.
D	Absolute language	The inner sizes must match.

3. Answer C Concept 2

An engineer at a water utility ~~serving 80,000 customers~~ must solve $Ax = b$ with $A = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$. ~~The data was collected over two dry summers.~~ What can she conclude?

Target word: solvability **Decides:** $\det A = 2 \cdot 2 - 4 \cdot 1 = 0$.

A	Inverted operation	Adds the diagonal products instead of subtracting them.
B	Absolute language	Nonzero entries do not guarantee invertibility.
C	Correct	$\det = 0$ means A has no inverse.
D	Confusion set	$x = 0$ as the only solution is the invertible case for $Ax = 0$; $\det = 0$ means none or infinitely many.

4. Answer D Concept 2 · transfer

For what value of k is $\begin{bmatrix} k & 3 \\ 4 & 6 \end{bmatrix}$ singular?

Target word: solve for k making $\det 0$ **Decides:** $6k - 12 = 0$.

A	Confusion set	Pairs the wrong entries: $4k - 18 = 0$.
B	Inverted operation	Sign error solving $6k - 12 = 0$.
C	Absolute language	With $k = 0$, $\det = -12$, which is not zero.
D	Correct	$6k = 12$.

5. Answer A Concept 3

A student at a college ~~known for its engineering co-op program~~ row-reduces an augmented matrix to $\begin{bmatrix} 1 & 2 & | & 3 \\ 0 & 0 & | & 5 \end{bmatrix}$. ~~The system came from a circuit with three resistors.~~ How many solutions does it have?

Target word: number of solutions **Decides:** Row 2 says $0 = 5$.

A	Correct	$0 = 5$ is a contradiction, so the system is inconsistent.
B	Camouflage	The resistor count does not decide consistency.
C	Adjacent category	That would need the row to read $0 = 0$.
D	Dropped element	Ignores the contradictory second row.

6. Answer B Concept 4

A data team at a retail chain ~~with 120 stores~~ builds a 5×8 feature matrix with rank 4. ~~The matrix was exported from a cloud warehouse.~~ What is the dimension of its null space?

Target word: nullity **Decides:** nullity = columns – rank = $8 - 4$.

A	Inverted operation	Adds rank to columns.
B	Correct	$8 - 4 = 4$.
C	Confusion set	Uses rows: $5 - 4$.
D	Dropped element	Ignores the rank.

7. Answer A Concept 4 · transfer

A 6×10 matrix has a null space of dimension 7. What is its rank?

Target word: solve for rank **Decides:** rank = columns – nullity = $10 - 7$.

A	Correct	$10 - 7 = 3$.
B	Inverted operation	Adds instead of subtracting.
C	Confusion set	Uses rows: $6 - 7$, which is impossible.
D	Wrong stem number	Repeats the nullity.

8. Answer C Concept 5

An analyst at a sports-analytics firm ~~with a famous client list~~ has four feature vectors, each with three entries, in $\mathbb{R}^3$. No vector is a multiple of another. ~~The data came from last season.~~ What must be true?

Target word: independence given count vs. dimension **Decides:** Four vectors in $\mathbb{R}^3$ exceed the dimension 3.

A	Adjacent category	A basis of $\mathbb{R}^3$ has exactly three independent vectors.
B	Dropped element	The count alone settles dependence here.
C	Correct	More than n vectors in $\mathbb{R}^n$ are always dependent.
D	Confusion set	Pairwise checks miss combinations of three or more vectors.

9. Answer B Concept 6

A graphics engine ~~used by an indie studio in Montreal~~ applies $T(x, y) = (2x, y + 1)$ to every vertex. ~~The studio's last game sold well.~~ Is T a linear transformation?

Target word: linearity test **Decides:** $T(0, 0) = (0, 1) \neq (0, 0)$.

A	Dropped element	Ignores the + 1 on y .
B	Correct	The + 1 is a translation, which breaks linearity.
C	Absolute language	Scaling by 2 alone is linear.
D	Confusion set	Affine is not the same as linear.

10. Answer D Concept 7

A vibration model for a wind turbine ~~installed offshore last year~~ uses $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$. ~~The turbine's blades are 60 m long.~~ What are A 's eigenvalues?

Target word: eigenvalues **Decides:** $\lambda^2 - 7\lambda + 10 = 0$ (trace 7, det 10).

A	Inverted operation	Sign error in the factoring.
B	Wrong stem number	The trace and determinant, not the eigenvalues.
C	Confusion set	Diagonal entries equal eigenvalues only for triangular matrices.
D	Correct	$(\lambda - 5)(\lambda - 2) = 0$; sum 7, product 10.

11. Answer A Concept 7 · transfer

A 3×3 matrix has eigenvalues 2, -1 , and 4. What is its determinant?

Target word: determinant from eigenvalues **Decides:** $\det = \text{product of eigenvalues}$.

A	Correct	$2 \times (-1) \times 4 = -8$.
B	Confusion set	The sum is the trace, not the determinant.
C	Absolute language	Only a zero eigenvalue makes it singular.
D	Inverted operation	Drops the negative sign.

12. Answer C Concept 8

An image-compression team ~~at a medical imaging vendor~~ computes singular values 10, 5, 2, and 1 for a scan matrix. ~~The scan came from a new machine.~~ If they keep only the top two, about what share of the matrix's total energy (sum of squared singular values) is retained?

Target word: energy retained by rank-2 approximation **Decides:** $(10^2 + 5^2) / (10^2 + 5^2 + 2^2 + 1^2) = 125 / 130$.

A	Adjacent category	Counts values kept (2 of 4), not energy.
B	Inverted operation	$5 / 130$ is the energy discarded.
C	Correct	$125 / 130 \approx 0.962$.
D	Dropped element	$15 / 18$ sums singular values without squaring.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Matrix product dimensions	$(m \times n)(n \times p)$ is defined only when the inner sizes match and gives $m \times p$; order matters.	Reporting the inner dimensions, or assuming AB and BA are the same size.	Two matrix sizes and a product.
2	Determinant and invertibility	A square matrix is invertible exactly when $\det \neq 0$; $\det = 0$ means no unique solution to $Ax = b$.	Thinking a matrix with all nonzero entries must be invertible.	A 2×2 or 3×3 matrix and a question about unique solutions or an inverse.
3	Consistency of linear systems	A row $[0 \dots 0 \mid c]$ with $c \neq 0$ means no solution; a zero row with free variables and no contradiction means infinitely many.	Reading a contradictory zero row as “infinitely many solutions.”	A row-reduced augmented matrix.
4	Rank-nullity	$\text{rank} + \text{nullity} = \text{number of columns}$.	Using the number of rows instead of columns.	A matrix size plus either its rank or its null-space dimension.
5	Linear independence and basis	More than n vectors in $\mathbb{R}^n$ are always dependent; a basis of $\mathbb{R}^n$ has exactly n independent vectors.	Checking only whether pairs of vectors are multiples.	A count of vectors compared with the dimension of the space.
6	Linear transformations	A linear map preserves addition and scalar multiplication, so it must send 0 to 0 ; adding a constant (translation) breaks linearity.	Calling any straight-line formula linear.	A formula with a “+ constant” term.
7	Eigenvalues, trace, and determinant	$Av = \lambda v$; eigenvalues come from $\det(A - \lambda I) = 0$, their sum is the trace and their product is the determinant.	Reading eigenvalues off the diagonal of a non-triangular matrix.	A small matrix or a list of eigenvalues.
8	SVD and low-rank approximation	Keeping the top k singular values retains $(\sigma_1^2 + \dots + \sigma_k^2) / \sum \sigma_i^2$ of the matrix’s total energy.	Summing singular values without squaring, or counting how many were kept.	A list of singular values and a “how much is retained” question.

Course scope (public description): This course develops foundational linear algebra knowledge and applies it to engineering and computing problems. Learners use matrix methods to formulate and solve systems of linear equations, analyze vector spaces and linear transformations to represent and manipulate multidimensional data, and determine eigenvalues, eigenvectors, and matrix factorizations to evaluate system behavior and optimize solutions. Emphasis is placed on applying mathematical principles to model, analyze, and solve real world problems, including applications in data representation, optimization, and the mathematical foundations of machine learning. The course builds on prior coursework in calculus for engineering and programming.